DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-ITCS112
COURSE TITLE	**:**	Fundamentals of Programming
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	2 units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	none
CO-REQUISITES	**:**	S-ITCS112LA Fundamentals of Programming Lab
PROFESSOR	**:**	MS. SHERRY B. NAZ
CONSULTATION HOURS	**:**	Mon 800 900 ONLINE Mon 1600 1700 COS100-C Tue 1400 1600 COS100-C Wed 1600 1700 ONLINE
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VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Introduces programming with emphasis on algorithmic and fundamentals programming of concepts. This also includes software development methodology, data types, control structures, arrays, and the mechanics of running, testing, and debugging. This course assumes computer literacy.

The course will look at solving problems using structured- programming approach. This class shall be delivered in both online and offline/module-based approach. This course is also

compliant with Outcomes-Based Education (OBE) and 21st Century Learning Design (21CLD) frameworks.

COURSE LEARNING OUTCOMES:

By the end of this course, students are expected to:

CLO1. Understand software methodology, algorithms, and flowchart.
CLO2. Create algorithms and flowcharts for solving a given problem.
CLO3. Understand common computer programming concepts.
CLO4. Analyze and evaluate pre-written programs.
CLO5. Develop programs to solve varieties of programming problems.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand software methodology, algorithms, and flowchart.	TLO1. Discuss software methodology, algorithms, and flowchart TLO2. Discuss flowcharting symbols and their uses
CLO2. Create algorithms and flowcharts for solving a given problem	TLO3. Draw and simulate flowcharts TLO4. Translate algorithms to flowcharts TLO5. Write algorithms for solving given problems as part of software development process.
CLO3. Understand common computer programming concepts.	TLO6. Learn syntax and semantics of Java programming language. TLO7. Understand identifiers, keywords, and operators in Java. TLO8. Explain I/O Statements
CLO6. Demonstrate 21st century skills in all learning activities	•*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessments/ Strategies**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**

Module 0	Week 1 (Aug. 19- Aug. 24)	• Gospel Reading, Reflection & Prayer • Presentation of **Module 0: Course Introduction** • Activity : *Meet the Students*	Schoolbook, PowerPoint, MS Forms	/		2
Module 1 CLO1, CLO2, TLO1 - TLO5	Week 2 (Aug. 26 - Aug. 31)	• Gospel Reading, Reflection & Prayer • Presentation of **Module 1: Algorithms and Flowcharting** • Activities: *Writing Algorithms, Drawing Flowcharts & Flowchart Simulation*	Schoolbook, PowerPoint	/		2
	Week 3 (Sept. 2 - Sept. 7)	•Gospel Reading, Reflection & Prayer •**Module 1 : Enabling Assessment -** Writing Algorithm and Drawing Flowchart, and/or Flowchart Simulation	Schoolbook, PowerPoint,		/	2

Module 2 CLO3, TLO6-TLO8	Week 4 (Sept. 9 - Sept. 14)	• Gospel Reading, Reflection & Prayer • Presentation of **Module 2: Introduction to Java Programming** • Activity : *Quiz on Programming Concepts*	Schoolbook, PowerPoint,	/		2
	Week 5 (Sept. 16-Sept. 21)	•Gospel Reading, Reflection & Prayer • Enabling Assessment : *Writing Progran Statenents*	Schoolbook, Java IDE & Compiler		/	2
Module 3 CLO4, TLO10-TLO12	Week 6 (Sept. 23 - Sept. 28)	• Gospel Reading, Reflection & Prayer	Schoolbook, PowerPoint, Java IDE and Compiler	/		2
		• Presentation of Module 3: Java's Selection Structure •Activity: Program Debugging and Tracing				
	Week 7 (Sept. 30 - Oct. 5)	• Gospel Reading, Reflection & Prayer			/	2
		Module 3: Enabling Assessment - *Program Debugging and Simulation*				
	Week 8 (Oct. 7 - Oct. 12)	Review / Consultation			/	2
Module 1 - 3	*Week 9 (Oct. 14 - Oct. 19)*	*Summative Assessment*		*/*		*2*

			TOTAL	***10***	***8***	***18***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Analyze and evaluate prewritten programs	TLO10. Debug and modify programs TLO11. Trace the output of the program
CLO5. Develop programs to solve varieties of programming problems.	TLO12. Write Java programs using the different Selection Structures. TLO13. Write programs using Java's Repetition Structure. TLO14. Perform basic manipulation with arrays
CLO6.Demonstrate 21st century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-world Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategy**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**
		• Gospel Reading, Reflection & Prayer				
Module 4 CLO4, CLO5, TLO10-TLO13	Week 10 (Oct. 21 - Oct. 26)	•Presentation of Module4: Repetition Structure Part 1 Activity: *Program Tracing and Debugging*	Schoolbook, PowerPoint, Java IDE and Compiler	/		2
	Week 11 (Oct, 28 - Nov. 2)	• Gospel Reading, Reflection & Prayer			/	2

		Independent Learning Week					
		• Gospel Reading, Reflection & Prayer					
	Week 12 (Nov. 4 - Nov. 9)	•Presentation of Module 4: Repetition Structure Part 2 Activity: *Program Tracing and Debugging*	Schoolbook, PowerPoint, Java IDE and Compiler	/		2	
		• Gospel Reading, Reflection & Prayer					
	Week 13 (Nov. 11 - Nov. 16)	Module 4: Enabling Assessment - *Programming, Tracing & Debugging*	Schoolbook, PowerPoint, Java IDE and Compiler		/		
		• Gospel Reading, Reflection & Prayer					
Module 5 CLO5, CLO6, TLO14	Week 14 (Nov. 18 - 23)	•Presentation of **Module 5: Arrays** • Activity: *Program Tracing and Debugging*	Schoolbook, PowerPoint, Java IDE and Compiler	/		2	

	Week 15 (Nov. 25 - Nov. 30)	• Enabling Assessment for Module 5 : *Programming, Tracing & Debugging*	Schoolbook, PowerPoint, Java IDE and Compiler		/	2
Module 4-5 CLO4-CLO6, TLO10-TLO14	Week 16 *Dec. 2 - Dec. 7)	• Gospel Reading, Reflection & Prayer Enabling Assessment: Problem Ser for Modules 4-5		/		2
	Week 17 (Dec. 9 - Dec. 14)	Review and/or Consultation			/	2
Module 4-5	*Week 18 (Dec. 16 - Dec.18)*	*Summative Assessment*		*/*		*2*
			TOTAL	***10***	***8***	***18***

GRADING SYSTEM:

Midterm

Class Participation	20%
Enabling Assessment	50%
Summative Assessment	30%
Total	***100%***

Finals

Class Participation	20%
Enabling Assessment	50%
Summative Assessment	30%
Total	***100%***

Overall Final Grade = Midterm + Final

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies

a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Onsite References

Provider	**Reference Material**
AEA Lib QA 76.73.J38 .F247a 2019	Farrell, J.c2019. Java programming:.8th edition. Philippine edition. (Taguig) : Cengage Learning Asia.
AEA Lib QA 76.73.J38 .Sch32 2019	Schildt, H .2019. Java : the complete reference.(New York): McGraw-Hill Education
AEA Lib QA 76.73.J38 .Sch35 2022	Schildt, H .2022.Java : a beginner's guide.9th edition.(New York): McGraw-Hill Education
AEA Lib QA 76.73.J38 .Sa94 2019	Savitch, W.c2019 Java : an introduction to problem solving & programming.8th edition.global edition.(New York): Pearson Education

Online References

Call number or e-provider	**Reference Material**
Website	Lean Java. [accessed 2024 July 30]. https://www.codecademy.com/learn/learn-java
Website	Java Tutorial. [accessed 2024 July 30]. https://www.w3schools.com/java/
Website	Learn Java Programming. [accessed 2024 July 30]. https://www.programiz.com/java-programming
Website	ProgrammingKnowledge. Java Tutorial for Beginners (Step By Step Tutorial). [accessed 2024 July 30]. https://m.youtube.com/playlist?list=PLS1QulWo1RIbfTjQvTdj8Y6yyq4R7g-Al

Prepared: **Information Technology Department**
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